

RISHAV RAJ

AI/ML Engineer | Python, Generative AI, Agents & MLOps

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Summary

AI/ML engineer with internship and project experience taking generative AI systems from fine-tuning and evaluation to API deployment and monitoring. Deployed a quantized Qwen2.5-Coder-7B model on AWS EC2, built tool-using LangGraph workflows, and implemented RAG with Pinecone. Python work includes modular services, automated tests, Docker, MLflow, and CloudWatch.

Technical Skills

Software Engineering: Python, SQL, FastAPI, REST APIs, modular design, automated testing, Git, GitHub Actions

LLMs & Agents: Hugging Face Transformers, prompt engineering, RAG, embeddings, Pinecone, LangGraph, MCP, PEFT/LoRA, TRL, GRPO

ML & MLOps: PyTorch, TensorFlow, MLflow, CloudWatch, execution-based evaluation, experiment design, adversarial testing

Cloud & Data: AWS EC2, AWS Lambda, Docker, MySQL, PM4Py, structured and unstructured data

Selected Projects

Text-to-SQL Fine-Tuning & Production Deployment

Mar – Jun 2026

github.com/dLounce/Text-to-SQL | [Live demo](#)

- Fine-tuned Qwen2.5-Coder-7B with LoRA SFT and GRPO using execution-based rewards, raising SQL execution accuracy from 50.6% to 68.4% on 1,600 questions from 21 databases excluded from training.
- Merged the SFT adapter and quantized it to GGUF Q4_K_M, cutting model size from about 15 GB to 4.4 GB while retaining 99% of SFT accuracy; served it with llama.cpp and FastAPI on AWS EC2 at about 12 seconds per CPU request.
- Added input validation, a hard timeout, health checks, and structured JSON logs; tracked four benchmark variants in MLflow and request count, errors, and latency in CloudWatch.

Sentinel Procurement | LangGraph Agent Security

Jun – Aug 2026

github.com/dLounce/sentinel-procurement

- Architected a LangGraph procurement workflow with isolated vendor-text interpretation, typed offer contracts, and deterministic budget, price, and delivery checks before order creation.
- Ran 100 live prompt-injection trials across 10 attack families; 12 changed the agent's structured decision beyond normal variation, but none caused an unauthorized order or policy violation.
- Covered parsing, policy enforcement, negotiation, and bypass attempts with 310 automated tests; a 25-cell ablation reproduced unauthorized orders in three deliberately weakened configurations.

Log2Agent | Event Logs to Agent Workflows

May – Jul 2026

github.com/dLounce/log2agent

- Built a Python pipeline that mined 36,796 BPI 2020 events with PM4Py and generated a LangGraph payment-request workflow with human approval gates for financial actions.
- Queried an MCP server to verify tool availability before code generation, then connected the generated agent to mock banking APIs.
- Replayed 50 historical cases through the workflow and matched every approval or rejection outcome; packaged the reproducible demo with Docker.

Experience

Artificial Intelligence Intern | Ballistic Learning Systems

Jun – Jul 2025

Faridabad, Haryana, India

- Built an evaluation set from 118 human-labelled answers and raised agreement between an LLM grader and human graders from 44.1% to 71.0% through prompt and retrieval changes.
- Developed a RAG pipeline for course PDFs by chunking unstructured documents, storing embeddings in Pinecone, and retrieving source context for grading and question generation.
- Designed a MySQL schema for quiz sessions, scores, and grading history.

Education

M.Sc. in Computer Science

Jul 2024 – Jul 2026

Central University of Rajasthan, Ajmer | CGPA: 7.1/10

Bachelor of Computer Applications

Aug 2018 – Jul 2021

Patliputra University, Patna | 73%

Certifications & Additional Information

UGC NET Computer Science, Qualified (Aug 2026) | Hugging Face Agents Course (2026)

AnalytiX DataOps Hackathon, CURaj (2025) | **Languages:** English, Hindi